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Palaeomagnetic results from southern Albania and their significance for the geodynamic evolution of the Dinarides, Albanides and Hellenides

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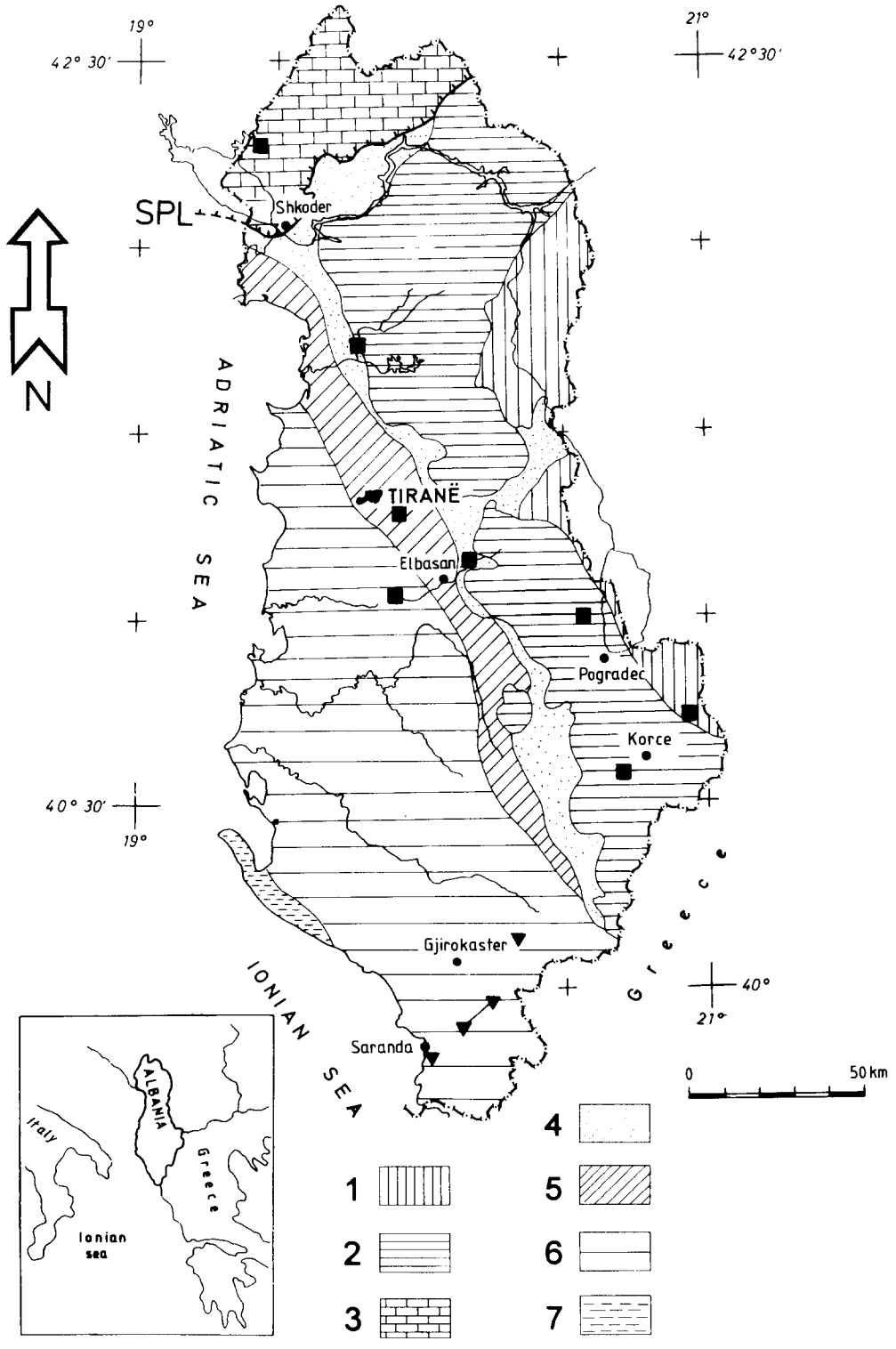
Abstract

Mesozoic and Cenozoic rocks were studied in three tectonic units in southern Albania. The area belongs to the Ionian Zone, Outer Albanides. Rock magnetic investigations, comprising alternating field and thermal cleaning, stepwise acquisition of isothermal remanent magnetization and hysteresis experiments were carried out. They established magnetite and titanomagnetite as the main carriers of the natural remanent magnetization. In the high field behaviour, haematite and goethite can also be seen. Several fold tests on the site and on a regional scale indicate that the directions are characteristic. These directions prove a general clockwise rotation of about 40° since the Tortonian. Using all available palaeomagnetic data from Albania, individual movements of the different tectonic units within the Inner and Outer Albanides can be deduced. This is also valid for the Albanian Alps, which are understood to be the southernmost part of the Dinarides.

1. Introduction

Since the early seventies a number of investigators have compiled palaeomagnetic data from the Mediterranean in order to describe and discriminate areas with unique rotation senses (Lowrie and Alvarez, 1975; Van den Berg et al., 1978; Channel et al., 1979; Lowrie, 1986; Westphal et al., 1986; Márton, 1987; Mauritsch and Becke, 1987). These papers suggest a central realm of counter-clockwise (CCW) rotations relative to the present North, called the Adriatic plate. However, in the Eastern Alps and in the transition zone from the Dinarides into the Albanides–Hellenides, abrupt changes of this regime have been recognised. Whereas in the Dinarides CCW rotations are reported (Márton and Veljovic, 1983; Márton, 1987; Márton and Milicevic, 1993), clockwise (CW) rotations were found in the Hellenides (Laj et al., 1982; Horner and Freeman, 1983; Kissel et al., 1985; Márton et al., 1990). In two investigations in Albania (Mauritsch et al., 1991; Speranza et al., 1992) the overall CW sense of rotation was demonstrated for the Albanides as well. The palaeomagnetic directions are of Upper Cretaceous and Cenozoic age and belong to the Ionian Zone, the Kruja Zone (the equivalent to the Gavrovo–Tripolitza Zone) and the Cukali–Krasta Zone (the equivalent of the Pindos Zone in Greece), as well as the

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19°
42° 30'
21°
42°30'
SPL
Shkoder
N
ADRIATIC SEA
TIRANË
Elbasan
Pogradec
Korce
40° 30'
19°
Gjirokaster
IONIAN SEA
Saranda
Greece
40°
21°
0
50 km
ALBANIA
Italy
Greece
Ionian sea
1
2
3
4
5
6
7

young overburden of the Mirdita and the Korab zones (Fig. 1). The age of the rotation coincides with observations made in Greece. The youngest CW rotation was found in the Inner Albanides in Upper Pliocene marls near Korca. This CW rotation can be seen throughout the Albanides and Hellenides, as well as in the Circum-Rhodope and the Serbo-Macedonian massifs (Kondopoulou, 1986; Pavlides et al., 1988; Edel et al., 1992). This general CW rotation is in good agreement with the sense and the age of rotations estimated from the geological reconstructions after Aubouin (1976) and Le Pichon and Angelier (1979). Taking into account all these results, the question arises, whether the Shkoder–Pec Line could be the transition zone between opposing senses of rotation.

In the Inner Albanides two phases (Upper Jurassic and Upper Cretaceous) of orogeny are reported, while in the Outer Albanides three phases are evident in Upper Liassic, Lower Miocene and Middle to Upper Miocene (Jacobshagen et al., 1978; Pumo et al., 1982).

The objective of this study was to complete the information about the geodynamic evolution of Albania presented by Mauritsch et al. (1991), which was based on northern and middle Albanian palaeomagnetic data of Mesozoic and Cenozoic rocks, sampled in 1989. About 500 samples were taken in the southern part of the Ionian Zone in Albania, covering the whole Mesozoic and the Tertiary until Oligocene. By using all available data from Albania (including unpublished data by Mauritsch et al.), special attention was paid to the problem of superimposed rotations and the significance of the observed directions for the geodynamic evolution of the Ionian Zone.

2. Geological setting and sampling

The area under investigation constitutes part of the Ionian Zone, a member of the external zone of the Albanides. In the western part, the Ionian Zone overthrusts the Preapulian foreland, whereas in the eastern part, it is overthrusted by the Kruja Zone and the Cukali–Krasta Zone. The tectonic features and the lithofacies development suggest that the Ionian Zone is a remnant of a miogeosynclinal area. Two major tectonic events, in the Lower Miocene and Middle to Upper Miocene, transformed the geosynclinal structure into an intensively folded zone. The Mesozoic comprises predominantly carbonates, overlying Permian–Lower Triassic evaporites. The sedimentation of terrigenous flysch deposits was initiated in the Oligocene and continued up to the Lower Miocene. These sediments are overlain by a Middle and Upper Miocene molasse formation of the pre-Adriatic depression. At the margins of the depression, the Tortonian rests on older formations with angular unconformity.

In the southernmost part of Albania, various nappes of the Outer Albanides contain carbonaceous sediments of Permo–Triassic to Quaternary age. The nappe and fold structure with an overall north-northwest (NNW) strike allows the application of palaeomagnetic fold tests on a local, as well as on a regional scale.

Seventeen localities were investigated during the sampling campaign in 1991, each comprising up to eleven sites to cover lithologic inhomogeneities and local structures. The geographic coordinates of the sampling sites and the stratigraphic age are given in Table 1. The majority of the localities are located along the road from Jorgucat across the Muzine Pass to Delvine (Fig. 2), representing a stratigraphic section from Eocene to Triassic age within a single tectonic unit, the *Mali i Gjere* structure. Several Palaeogene sites and a Cretaceous site were sampled by Speranza et al. (1992) along this section. In order to identify possible separate rotations of the nappes, a few Lower and Upper Cretaceous sites in the Ionian tectonic zone near Saranda as well as a Palaeogene locality in the upper tectonic unit

Fig. 1. Simplified geologic map of Albania (after Pumo et al., 1982). Inner Albanides: *1* = Korab Zone; *2* = Mirdita Zone. Albanian Alps: *3* = Outer Albanides; *4* = Cukali–Krasta Zone; *5* = Kruja Zone; *6* = Ionian Zone; *7* = Sazan Zone. *SPL* = Shkoder–Pec Line. Squares show localities sampled in 1989 (Mauritsch et al., 1991), triangles localities for this study (see Fig. 2 for details).

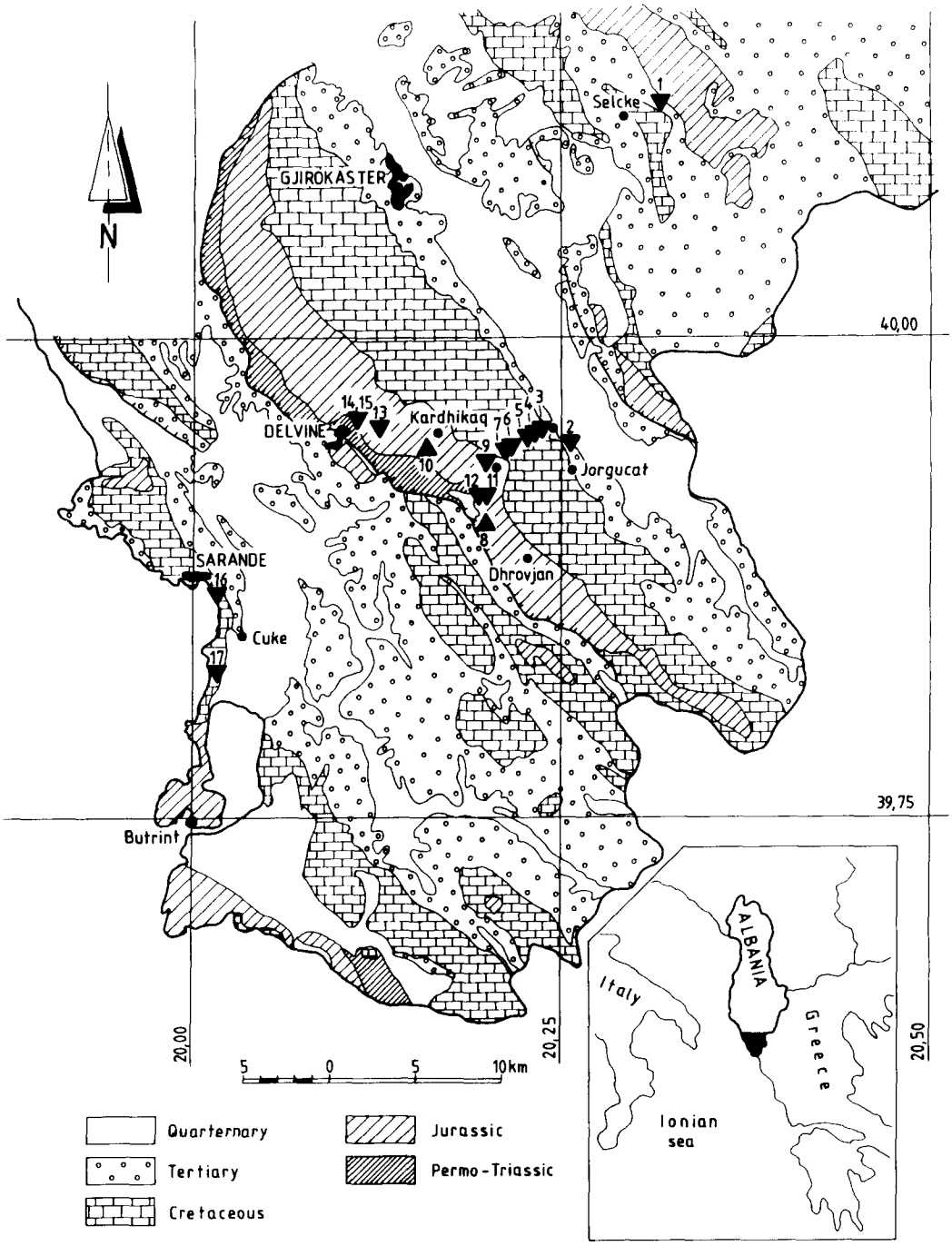


Fig. 2. Simplified geologic map of the sampling area, the southernmost part of Albania (after Geologic Map of PSR of Albania, 1:200,000, Tirana 1983). Triangles mark the sampling localities for this study.

Table 1
Location and stratigraphic age of southern Albanian sampling sites

Locality Nr.	Location	Sites	Lat. °N	Long. °E	Stratigr. age
1	Selcke	G 1–G 6	40.116	20.323	Paleogene
2	Jorgucat	G 15–G 25	39.940	20.257	Eocene
3	Grapsh	G 26–G 30	39.947	20.238	Maastrichtian
4	Grapsh	G 31–G 35	39.945	20.233	Campanian
5	Grapsh	G 36–G 41	39.942	20.229	Coniacian
6	Muzine	G 42–G 43	39.937	20.217	Neocomian
7	Muzine	G 44–G 50	39.936	20.212	Tithonian
8	Kardhikaq	G 74–G 76	39.906	20.198	Middle Jurassic
9	Muzine	G 54–G 59	39.929	20.200	Middle Jurassic
10	Kardhikaq	G 77–G 80	39.947	20.158	Lower/Middle Jurassic
11	Dhrovjan	G 69–G 73	39.913	20.199	Lower/Middle Jurassic
12	Syrikalter	G 60–G 68	39.913	20.195	Lower Jurassic
13	Delvine	G 109–G 112	39.949	20.116	Lower Jurassic
14	Delvine	G 113–G 115	39.952	20.112	Upper Triassic
15	Delvine	G 116–G 119	39.951	20.111	Triassic
16	Sarande	G 81–G 83	39.860	20.020	Upper Cretaceous
17	Cuke	G 84–G 88	39.820	20.019	Lower Cretaceous

near Selcke, the northernmost location of the study area (Fig. 2), were sampled.

Locality 2 (Eocene) consists of sandy flysch including foraminifera-rich layers and limestones. Upper and Lower Cretaceous limestones were sampled in localities 3 to 6 including a fold test in the Maastrichtian. The Neocomian locality near the village Muzine shows evidence of intense folding. On the western flank of the Mali i Gjere structure seven Jurassic and two Triassic localities were chosen. The Upper Jurassic limestones are interbedded with dolomites and radiolarian cherts (locality 7), the Middle and Lower Jurassic consists of oolitic limestones, platy radiolarian limestones with filaments and radiolarian cherts (localities 8–13). The Triassic sampled in the vicinity of Delvine (localities 14, 15) is characterized by stromatolitic limestones and limestones with megalodonts.

The sampling was carried out by core drilling and orientation with standard palaeomagnetic techniques, except for the three localities near Delvine, where a number of oriented hand samples were taken. All together a set of 456 cores and 51 hand samples were collected.

The bedding planes scatter around a 60° declination of azimuth with varying inclinations, corresponding to the regional structures (Fig. 3). The overall fold axis, obtained by analysis of these planes, is subhorizontal with a NNW alignment, thus enabling the use of simple tilt correction for the palaeomagnetic data.

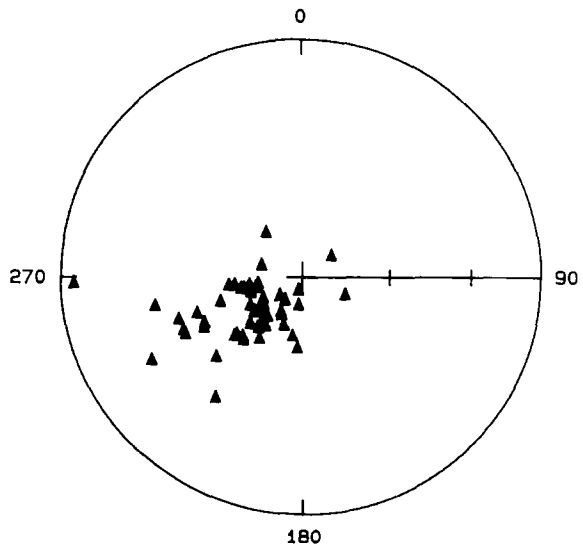


Fig. 3. Bedding poles of southern Albanian sampling sites from the Upper Triassic to the Palaeogene. Equal-area projection.

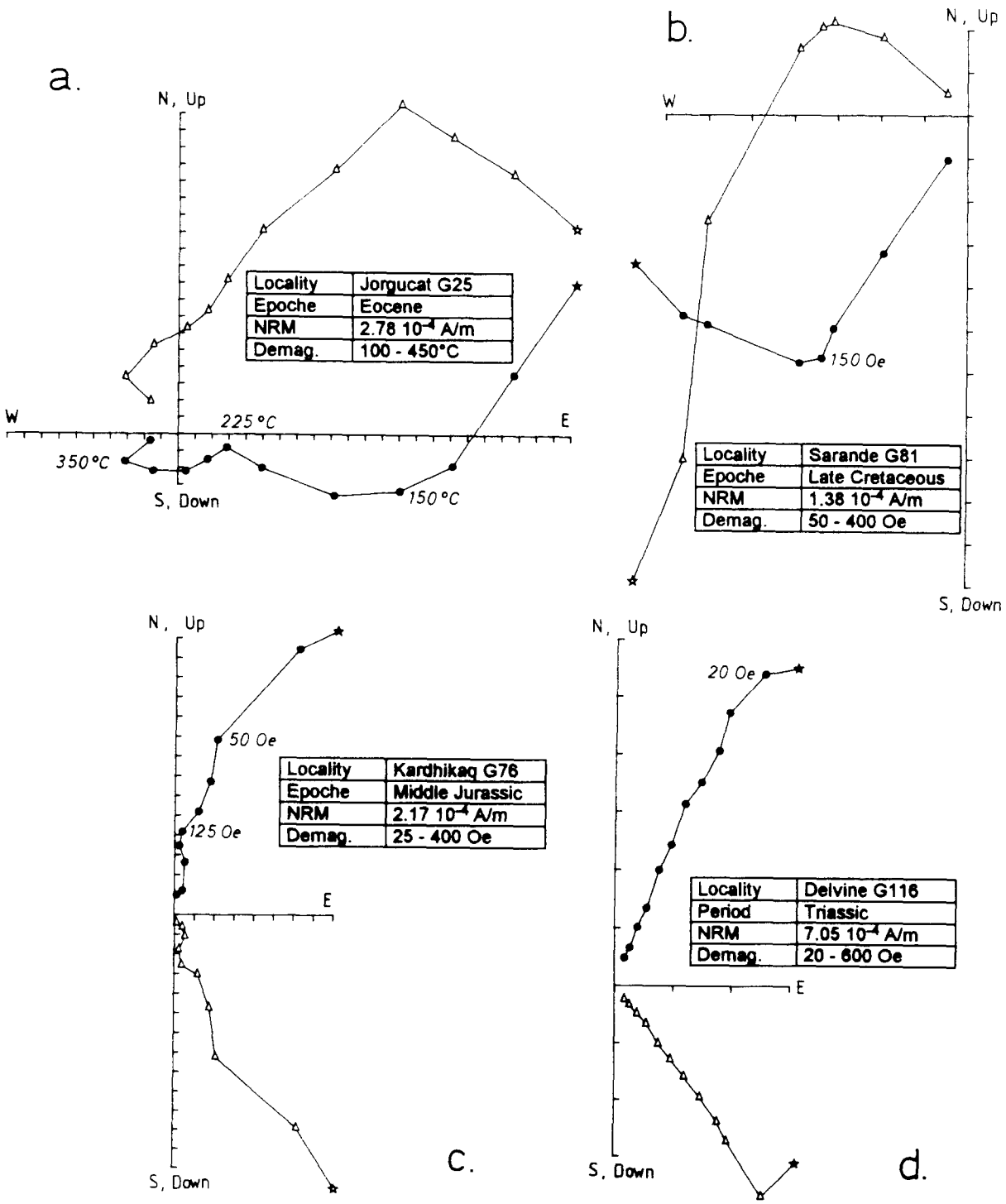


Fig. 4. Thermal and alternating-field demagnetization diagrams in orthogonal projection. Tilt-corrected. Circles and triangles represent projection on the horizontal and vertical planes.

3. Rock magnetic investigations

Within each site, two or more samples were chosen for either thermal or alternating field demagnetization, in order to study the stability of different components of the natural remanent magnetization (NRM). Additionally, the isothermal remanent magnetization (IRM) was observed before and after thermal treatment. All measurements were carried out in the palaeomagnetic laboratory of the Mining University in Gams using a 2G three-axis cryogenic magnetometer.

The demagnetization paths of both thermal and alternating field treatments show a single stable characteristic remanence vector after a few cleaning steps (Fig. 4) in most of the samples, whereas in some samples a demagnetization plane with two antipodal directions can be obtained (Fig. 4a). This was also described by Speranza et al. (1992). By cleaning additional samples from the same site, a palaeomagnetic direction can then be calculated as the intersection of the demagnetization planes, presented as great circles in an equal-area projection (Fig. 5). These calculated directions are in good agreement with the stable end directions for equivalent samples after the different demagnetization methods. An exception are the Triassic localities, where just stable end directions appear; in all other localities both behaviours occur.

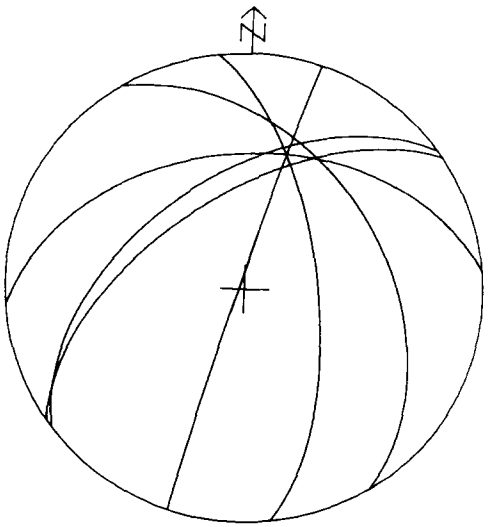


Fig. 5. Demagnetization planes obtained from six Lower Jurassic samples. Equal-area projection.

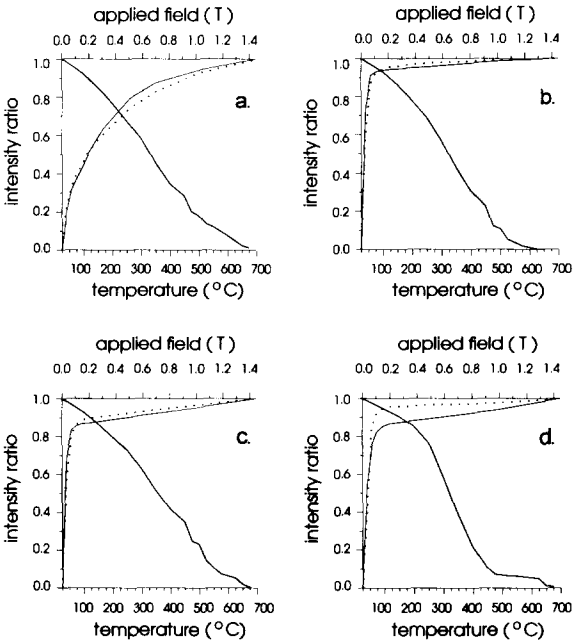


Fig. 6. Stepwise acquisition of isothermal remanent magnetization (IRM) and subsequent thermal demagnetization of IRM for samples of different ages. (a) Palaeogene, locality 1. (b) Coniacian, locality 5. (c) Tithonian, locality 7. (d) Triassic, locality 15. Dotted curves represent IRM after thermal treatment at 150°C or higher.

Stepwise acquisition of IRM and subsequent thermal demagnetization (Is–T curves) give evidence of a magnetite and/or titanomagnetite dominated magnetic phase in the majority of the samples (Fig. 6). Both magnetite and haematite can be visualized in the Is–T curves. A minor influence of goethite can be demonstrated by comparing the acquisition behaviour before and after heating the samples (Fig. 6d). This is particularly pronounced in the Upper Triassic.

Measurements of low field susceptibility produced a weak, predominantly diamagnetic bulk susceptibility for the majority of the samples, indicating a limited "ferromagnetic" content in the mineral assemblage. This corresponds with the weak NRM intensities in the range of 2×10^{-4} A/m.

4. Palaeomagnetic investigations

All samples of Palaeogene age were thermally demagnetized progressively, the Mesozoic samples were selected individually for thermal or alternating field treatment based on natural remanent intensity and mechanical condition (some of the micritic limestones exploded during heating at about 300°C). The characteristic remanent magnetization (ChRM) directions for single samples were calculated using the standard least-squares method (Kirschvink, 1980). Mean values for sites and localities were calculated following Fisher (1953).

Two fold tests were possible in the Palaeogene of Selcke and one in the Maastrichtian of Grapsh. All of them pass the statistical tests at a 95% significance level (McElhinny, 1964; McFadden and Jones, 1981; McFadden, 1990). Within localities of diverse bedding, the scatter of corresponding site mean directions also decreases significantly after tilt correction (Fig. 7, Table 2).

The Palaeogene locality near Selcke shows several indicators for the reliability of the palaeomagnetic data: two positive fold tests (visualized in Fig. 7a by a decrease of the confidence cones of the site mean directions), a distinct decrease of the overall confidence cone of the locality (α_{95} = 22.1° before and 8.6° after correction; Table 2; Fig. 8a) and the occurrence of a perfectly antipodal direction after tilt correction (Fig. 7a). Because of the uniform bedding pattern, the effect of tilt correction is smaller in the Eocene locality of the second tectonic unit (Fig. 7b). Here the reverse polarity is dominant, but antipodal normal polarity also occurs. The mean directions in the two Palaeogene localities, which are in different tectonic units, indicate a clockwise rotation of 39° and 43° relative to the recent magnetic north direction (Table 2, Fig. 8a). The overall mean for the two localities gives a palaeomagnetic direction for the Palaeogene age group of $D = 42°$ and

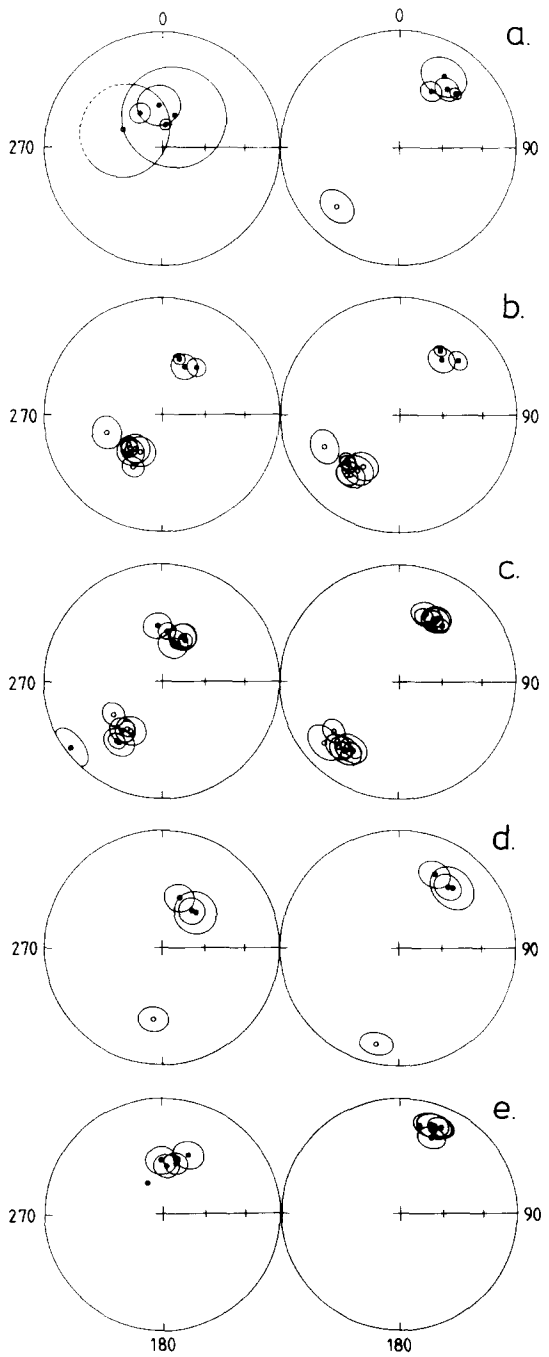


Fig. 7. Site mean directions with α_{95} confidence circles before (left) and after (right) tilt correction. Equal-area projection with dots for lower hemisphere and open symbols for upper hemisphere. (a) Locality 1 (Palaeogene) including two fold tests. (b) Locality 2 (Eocene). (c) Localities 3–5 (Maastrichtian, Campanian, Coniacian) including a fold test. (d) Locality 7 (Tithonian). (e) Localities 14–15 (Upper Triassic).

$I = 37°$, defined by fifteen sites with an $\alpha_{95} = 4.4°$ (Table 2).

The Upper Cretaceous localities 3 to 5 on the eastern flank of the Mali i Gjere structure produce the same results for Maastrichtian, Campanian and Coniacian palaeomagnetic directions, with evidence of clockwise rotation, too (Fig 7c). The data are supported by a positive fold test in the Maastrichtian. These localities can be combined with the Upper Cretaceous in the southernmost tectonic unit near Sarande (locality 16) to allow a clearly positive fold test on a regional scale (Fig. 8b). Consequently, we combine the results of the time span from Maastrichtian to Coniacian, leading to a palaeomagnetic direction for the Upper Cretaceous of $D = 35°$ and $I = 33°$, based on seventeen sites with $\alpha_{95} = 4.4°$ (Table 2). Following the data presented above, no significant rotations took place between the Upper Cretaceous and Palaeogene.

Lower Cretaceous data differ substantially from those presented above. The Neocomian samples, collected at the Muzine pass in a strongly tectonized site, displayed well defined remagnetization circles. Their subparallel orientation does not allow the computation of a statistically significant direction but rather defines an average plane. By combination of principal component analyses with the remagnetization planes, a site mean of six samples with an $\alpha_{95} = 13.0°$ was obtained. In contrast to this, the Lower Cretaceous of Cuke (locality 17) contained mainly stable characteristic magnetization components (Fig. 4b), but produced comparatively scattered directions, resulting in an α_{95} confidence cone of 22.2° for the locality mean (Table 2). While this direction (D

Table 2
Locality mean directions and age group mean directions for southern Albanian sites

Local. Nr.	Age	N (S)	nor/rev	D/I bc	α_{95}	K	D/I ac	α_{95}	K
1	Palaeogene	5 (26)	4/1	14/73	22.1	12.9	39/34	8.6	80.1
2	Eocene	10 (53)	3/7	218/−55	6.2	60.7	223/−38	5.5	76.6
3	Maastrichtian	3 (16)	2/1	303/64	32.3	1.9	34/33	26.2	23.3
4	Campanian	5 (24)	0/5	219/−45	8.8	75.9	219/−30	6.5	138.4
5	Coniacian	6 (32)	5/1	25/56	6.9	95.0	32/38	7.1	87.9
6	Neocomian	1 (6)	0/1	141/−31			156/−23		
7	Tithonian	4 (21)	3/1	25/52	16.2	33.1	29/29	15.9	34.4
8	JM	3 (16)	3/0	349/45	21.8	32.9	8/41	21.8	32.9
9	JM	4 (17)	4/0	344/55	12.7	53.2	4/50	10.9	71.8
10	JL/JM	4 (22)	4/0	15/48	2.9	1000.5	18/38	2.5	1358.3
11	JL/JM	3 (5)	3/0	3/46	g.c.		16/33	g.c.	
12	JL	6 (16)	6/0	347/51	10.6	40.9	16/41	10.4	42.5
13	JL	4 (9)	4/0	34/56	9.4	96.5	36/45	9.7	90.3
14	TRU	3 (11)	3/0	359/58	20.7	38.8	19/21	10.0	153.2
15	TR	3 (17)	3/0	15/50	13.0	90.7	21/25	7.9	250.0
16	KU	3 (17)	0/3	220/−17	5.5	509.1	217/−28	7.4	277.4
17	KL	3 (11)	3/0	15/40	22.2	32.0	3/36	22.2	32.0

Age group mean directions:

Age group	N	D/I bc	α_{95}	K	D/I ac	α_{95}	K	Pole lat. °N/long. °E
Palaeogene	15	33/61	8.5	21.3	42/37	4.4	77.3	49/126
Upper Cretaceous (KU)	17	33/43	10.1	13.6	35/33	4.4	66.6	53/136
Middle Jurassic (JM)	7	346/50	9.5	41.3	6/46	8.9	47.0	76/177
Lower/Middle Jurassic (JL/JM)	4	13/48	4.2	336.5	18/37	2.9	678.4	65/156
Lower Jurassic (JL)	10	4/55	10.8	20.9	26/42	7.4	43.4	63/138
Upper Triassic (TRU)	6	8/54	10.0	46.2	20/23	4.8	198.9	57/162

N (S) is number of sites (samples); nor/rev is number of sites in normal/reverse polarity; D/I bc is characteristic remanence direction before tilt correction; D/I ac is after tilt correction; g.c. is great cirle intersection.

= 3°, $I = 36°$) fits better into the general picture, the Neocomian of Muzine appears to be influenced by local tectonics.

The Jurassic data are of varying quality, but are homogeneous in their palaeomagnetic setting. The majority of the directions observed are of normal polarity, reversed once only in the Tithonian sites (Fig. 7d).

The locality mean directions show a discrimination throughout the Jurassic (Table 2; Fig. 8c). Besides the above-mentioned Tithonian, a Middle Jurassic mean direction of $D = 6°$ and $I = 46°$, a Lower/Middle Jurassic boundary direction of $D = 18°$ and $I = 37°$, and a Lower Jurassic direction of $D = 26°$ and $I = 42°$ can be distinguished (data from two localities each; Table 2). The discrimination is significant at a better than 99% confidence level, tested following both Watson (1956) and Mardia (1972).

Triassic samples are characterized by simple demagnetization paths and well defined remanence directions within both localities (Fig. 7e). The scatter of the mean directions decreases significantly after tilt correction (Fig. 8d). The Upper Triassic palaeomagnetic mean direction from two localities, containing six sites is $D = 20°$ and $I = 23°$ with $\alpha_{95} = 4.8°$ (Table 2).

5. Conclusions

The palaeomagnetic data presented above suggest pre-folding magnetization directions, proved by successful laboratory experiments and field tests. At several sites fold tests were carried out, antipodal directions were found, and results from three different tectonic units were found to be in the same range.

The data set from Albania covers a continuous time span from Upper Triassic to Tortonian (Table 3). Reference directions are the predicted values for the present geographic position of Al-

Table 3
Compiled paleomagnetic data from Albania

Stratigraphic age	Age (Ma)	Rock unit	N	D/I ac	α_{95}	Reference (local. Nr.)
Tortonian	7–10	Outer Albanides	7	32/40	7.7	A
Tortonian	7–10	Inner Albanides	3	40/38	10.9	A
Chattian	23–29	Inner Albanides	11	40/44	8.6	A
Oligocene	23–35	Outer Albanides	3	42/28	29.4	C
Palaeogene	23–65	Outer Albanides	5	39/34	8.6	B (1)
Eocene	35–56	Inner Albanides	5	57/43	19.5	A
Eocene	35–56	Outer Albanides	10	43/38	5.5	B (2)
Eocene	35–56	Outer Albanides	11	43/39	4.0	C
Maastrichtian	65–74	Outer Albanides	14	36/43	8.7	A
Maastrichtian	65–74	Outer Albanides	3	34/33	26.2	B (3)
Upper Cretaceous	65–97	Outer Albanides	5	49/41	10.1	A
Upper Cretaceous	65–97	Outer Albanides	3	37/28	7.4	B (16)
Campanian	74–83	Outer Albanides	5	39/30	6.5	B (4)
Coniacian	87–89	Outer Albanides	6	32/38	7.1	B (5)
Cenomanian	90–97	Inner Albanides	2	43/22	11.3	A
Lower Cretaceous	97–146	Albanian Alps	7	51/35	9.9	A
Lower Cretaceous	97–146	Outer Albanides	3	3/36	22.2	B (17)
Tithonian	146–152	Inner Albanides	7	50/41	7.0	A
Tithonian	146–152	Outer Albanides	4	29/29	15.9	B (7)
Middle Jurassic	157–178	Outer Albanides	7	6/46	8.9	B (8, 9)
Lower/Middle Jurassic	178	Outer Albanides	4	18/37	2.9	B (10, 11)
Lower Jurassic	178–208	Outer Albanides	10	26/42	7.4	B (12, 13)
Upper Triassic	208–235	Outer Albanides	6	20/23	4.8	B (14, 15)

Ages taken from Harland et al. (1990), rounded. N is number of sites. D/I ac is characteristic remanence direction after tilt correction. References are: (1) Mauritsch et al. (unpublished); (B) this paper; (C) Speranza et al. (1992).

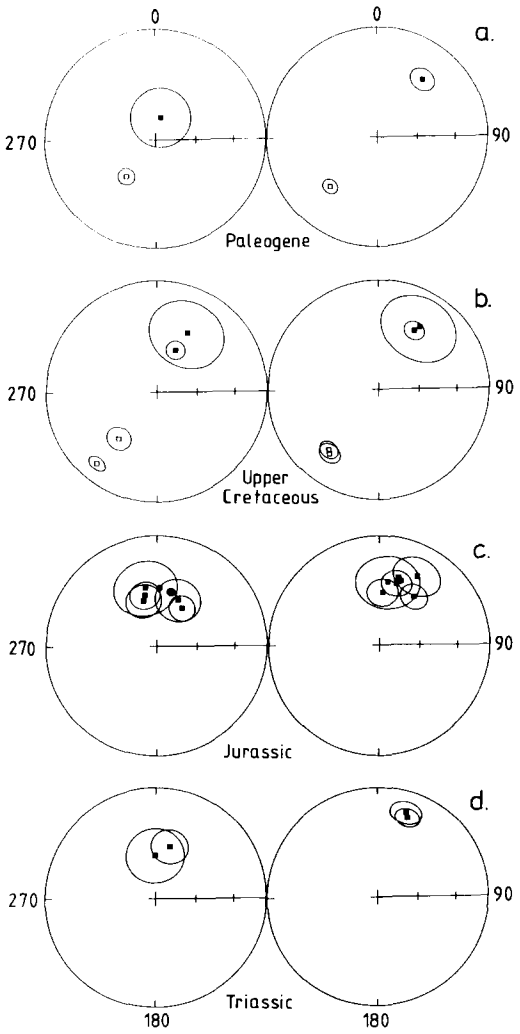


Fig. 8. Locality mean directions with α_{95} confidence circles before (left) and after (right) tilt correction. Equal-area projection with dots for lower hemisphere and open symbols for upper hemisphere; (a), (b) and (d) show concentration within the Palaeogene, Upper Cretaceous, and Triassic age groups, respectively, and (c) in the Jurassic.

bania (latitude 40°N; longitude 20°E), calculated from African and Eurasian *master apparent polar wander paths* (200–10 My) after Besse and Courtillot (1991). The comparison demonstrates a clockwise rotation of the Outer and Inner Albanides, which must be dated as younger than Tortonian (Fig. 9a).

Elimination of this tectonic movement by back-rotating the whole data set 40° counterclockwise, produces a close correlation with the derived curves (Fig. 9b). In particular, a strong affinity with the African curve is evident for Upper Triassic up to Lower Cretaceous results. A Neocomian result from the Albanian Alps, i.e., north of the Shkoder–Pec Line, shows affinity to Eurasian directions, which seems to be due to local tectonics which occurred as a consequence of transversal movements along this line. A comparison of the palaeodeclinations from the Ionian Zone and the overlying tectonic units shows a systematic CW rotation of the latter (Fig. 9). For the Upper Cretaceous and the Tertiary, a perfect overlap (95% confidence limit) of the expected palaeodirections can be seen, with only a slight shallowing in the inclination values in Eocene and younger rocks (Fig. 9b).

The Albanian palaeodirections presented above are compatible with the Mio–Pliocene results from the island of Corfu (Laj et al., 1982). Thus, combining these data sets provides a continuous and detailed palaeomagnetic record from Upper Triassic to recent times. The overall CW rotational movement can be dated from 5 million years to the present. This clockwise rotation can be found on the island of Corfu (Laj et al., 1982), in the Inner and Outer Albanides (Mauritsch et al., 1991; Speranza et al., 1992; this paper), Inner and Outer Hellenides (Horner and Freeman, 1983; Kissel et al., 1986; Márton et al., 1990), Chalkidiki ophiolites (Edel et al., 1990) and the Serbo-Macedonian massif (Pavlides et al., 1988). In the south and southeast of the Hellenic arch, no rotation was identified (Laj et al., 1982).

In contrast to this, the Dinarides, as far as they were investigated, show counterclockwise rotation in agreement with the Adriatic realm (Márton and Mauritsch, 1990; Márton and Milicevic, 1993).

Summing up all available results, it can be said that the Shkoder–Pec Line does not seem to be the transition zone between the opposing senses of rotation. Since the southernmost results in the Dinarides (from the island of Vis and Dugi Otok, west of the Split Line) show a clear dominance of counterclockwise rotation (Márton and Milicevic, 1993), and the northernmost results of Albania

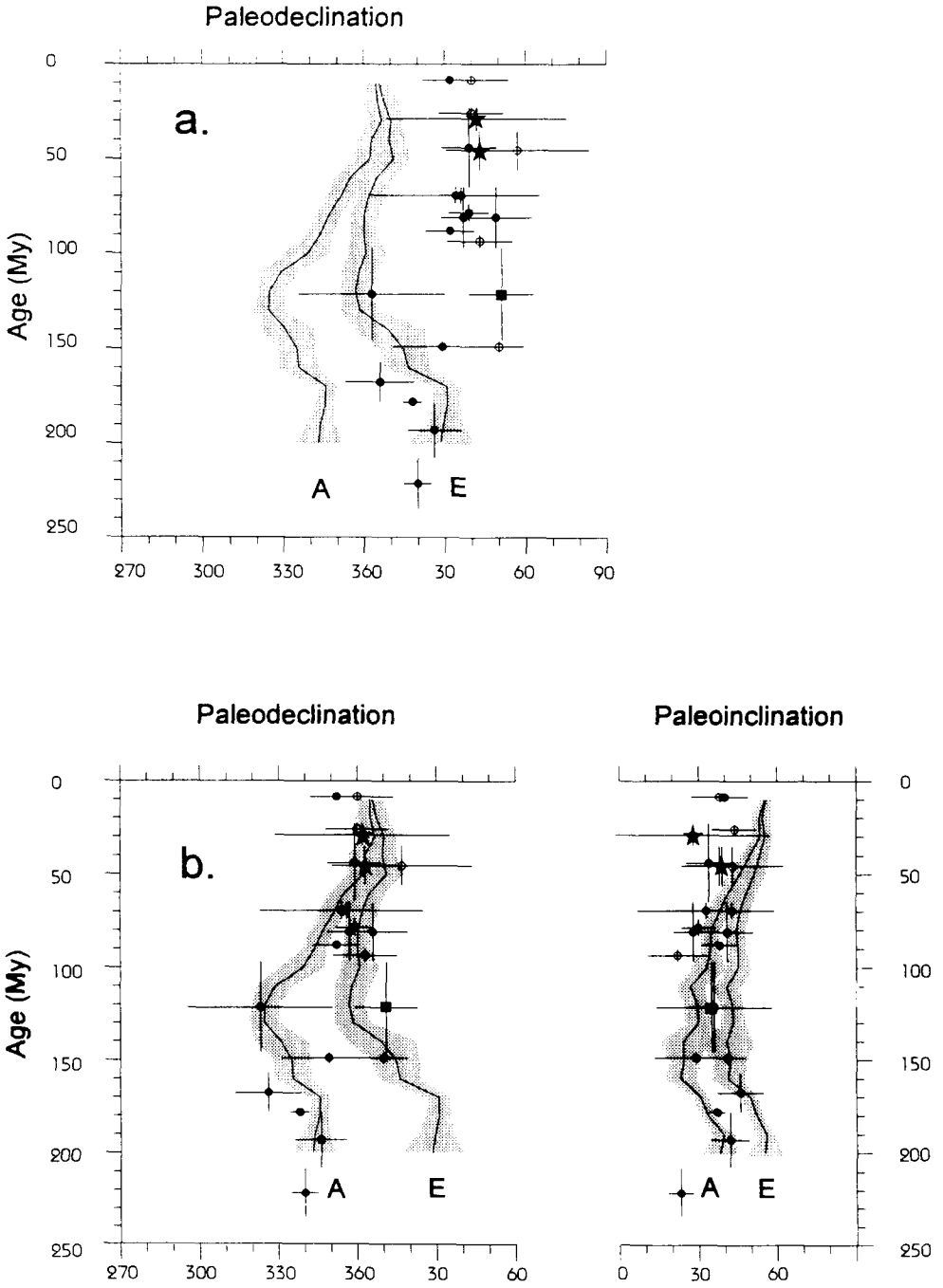


Fig. 9. Compiled palaeomagnetic data from Albania and predicted palaeodirections for Africa (*A*) and Eurasia (*E*), calculated from *master apparent polar wander paths* (Besse and Courtillot, 1991) for a location 40°N, 20°E with their associated 95% confidence limits (shaded). (a) Albanian data as listed in Table 3. (b) Albanian data after 40° counterclockwise rotation. Dots for Inner Albanides and circles for Outer Albanides, square for Albanian Alps. Asterisks represent data taken from Speranza et al. (1992).

(north of the Shkoder–Pec Line) still show clockwise rotation, the solution for the diverging rotation senses could be found in block rotations between the Split–Sarajewo Line and the Shkoder–Pec Line. This would fit the concept of transverse structures in the external and internal zones of the Dinarides and Hellenides (Aubouin, 1976). The clockwise rotation is in good accordance to the reconstruction of Le Pichon and Angelier (1979). However, this model has to be slightly modified in the westernmost edge to accommodate the uniform clockwise rotation recognised up to the north of the Shkoder–Pec Line.

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